

DGMD E57 Mini Design Challenge III: Multimodal Creation

Mini Design Challenge III Process Document

For this assignment, I expanded on Mini Design Challenge II by create a 1-page handout for students as a guide for their project. This corresponds to the objective for students to self-assess their progress toward the learning goals and the success criteria of the PBL project. Students would individually have a copy of this document, in print and/or digital form. They would use it to formatively assess their progress at each of the four project phases. It could also be used for peer assessment at the conclusion of the project. It has three main sections: 1) Learning Objectives that represent different levels of Bloom's Taxonomy and different skills. These objectives were written with cross-curricular standards in mind; 2) Assessment Criteria that students use to check their progress individually and as a group. Teachers could direct students to use it as a checklist or formulate a rating or number system (better still, students could be in charge of this system); and 3) Standards. These are pulled directly from standards documents and represent the main assessed standards for this project. There are other supporting standards that the teacher can formatively address throughout.

I used the following Learning Design principles to guide the creation of this multimedia artifact:

- PBL/experiential learning: learners take part in a realistic project and produce an actual artifact by its end. They are given realistic constraints and criteria to follow. They participate in several rounds of self and peer assessment to guide their designs.
- Cognitive load: This artifact reduces cognitive load by providing a visual representation and reminder of project objective and assessment criteria. Students use the artifact to check their progress formatively, and often.
- Backwards design: Standards and assessments were determined before learning objectives and activities were planned. This allowed for clearer alignment between the project goals and student tasks. This document connects the learning activities (objectives) with standards and assessment.
- Learning in context: addressed through PBL, students develop skills and knowledge by working through a hands-on project set in a real-world scenario. This artifact keeps learners within the context of the project.

AI collaboration process:

I utilized NotebookLM which contains my project files, including all standards documents, my project brief, and PBL guidance. NotebookLM allows for source-specific generation and allows for limited source generation, which can cut down on errors and hallucinations. It also keeps all of my uploaded files private and sandboxed. Using the standards documents for social studies, math, science, and ELA, and my Project Brief, I asked it to create a 1-page infographic document with some guidance (see below for my prompt). I generated an image on Standard mode at first for the first pass (assuming this uses Nano Banana but not entirely sure), then two iterations on the Detailed (beta) mode. Each iteration, I made adjustments to my prompt based on the results.

Results were overall what I was hoping for, with a couple caveats. One is inconsistent spelling and grammar due to generative text mistakes. The other are accessibility issues, such as color contrasts, font consistencies and graphics that are generally too "busy." Here is how I could address these issues:

- Use NotebookLM for an outline of an infographic then make it manually using a design tool such as Adobe Illustrator or Figma.
- Use generative AI to produce individual assets, such as graphics, icons, tables, etc. Then manually create text-based assets and combine the two, again in a design tool.
- Use a different AI model or tool that is known to be better at generating text and addressing accessibility concerns.
- Similarly, I could run the generated artifact through a different image generation, using image-to-image generation to correct spelling or make other minor changes.